G. Leigh Harwell

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Academic Website
harwell.gwen@gmail.com

Education:

University of Oklahoma — August 2023 – May 2027

Bachelor of Science in Astrophysics | Minor in Mathematics | Data Analytics Certificate

Norman, OK, USA

Research Experience:

Multi-Messenger Observations of Compact Objects — August, 2026 – Present

University of Oklahoma, Department of Physics and Astronomy — *Norman, OK, USA*

- Characterizing double white dwarf systems through multi messenger observations.
- Analyze multi messenger datasets to identify compact binary candidates for follow-up observational studies.

Machine Learning and Stellar Parameter Estimations — October, 2024 – October 2026

University of Oklahoma, Department of Physics and Astronomy — *Norman, OK, USA*

- Transitioned project into an intensive NSF REU at the University of Oklahoma during Summer 2026, focusing on advanced stellar parameter estimation.
- Apply machine learning techniques to asteroseismology and galactic archaeology datasets to improve stellar classification models.
- Implementing "TheCannon" data driven model to estimate stellar parameters in Astroseismolog with elemental abundances from spectroscopic data

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Other Experience:

Physics for Life Science Majors Tutoring Coordinator & PLA — August 2025 — Present

Action Tutoring | Student Learning Center | University of Oklahoma — *Norman, OK, USA*

- Coordinating the tutoring for the life science majors, and collaborating with department faculty to provide extra problem sets and conceptual worksheets for exam review.
- Provide individual and small-group tutoring sessions in introductory mechanics, optics, thermodynamics, and electromagnetism.

Research Presentations:

Characterizing Stellar Labels with Machine Learning 2026

Department Research Symposium, University of Oklahoma *Norman, OK, USA*

Harwell, G. L., Beaty, M., Hayden

- Poster presentation detailing the findings from the UReCA Fellowship program over my research on machine learning and stellar parameter estimations.

Multi-Messenger Observations of Compact Binary Systems 2026

Capstone Research Presentation, University of Oklahoma *Norman, OK, USA*

Harwell, G. L., Barrientos, M., Kilic, M.

- Five minute oral presentation over the background for my capstone research on multi-messenger observations.

Awards & Honors:

UReCA Fellowship 2026

The University of Oklahoma *Norman, OK, USA*

- Funded collaborative research developing machine learning tools for stellar label estimation; presented findings at department symposium.

Physics and Astronomy Meritorious Scholarship 2024 - present

Homer L. Dodge Department | University of Oklahoma *Norman, OK, USA*

- Awardees to outstanding physics and astronomy students that maintain a 4.0 GPA throughout the semester.

President's Honor Roll 2024

The University of Oklahoma *Norman, OK, USA*

- Award to distinguished students that maintain a 4.0 GPA throughout the semester.

Skills:

Programming Languages: Python (intermediate) (Scientific Libraries: NumPy, SciPy, Pandas, Matplotlib, AstroPy, and scikit-learn) | LaTeX (beginner)

Data Analysis: Multi-messenger data analysis | stellar parameter modeling | machine learning workflow

Computer Literacy: Utilize various tools in the operation of computers and various other technologies.

Adaptability: Respond flexibly to evolving factors, environments, and conditions by using new skills.

Extracurriculars:

Community and Belonging August 2026 — Present

Society of Physics Students August 2023 — Present

Women in Physics August 2023 — Present